Review changelog: batch p0p1 (Phase 0 lessons 01-21, Phase 1 lessons 01-20)

Review changelog: p0p1 part A (00-web-and-javascript-foundations 01-07), 2026-10-22

Methods: node 24 (strip-types) run of all ts/js code blocks in 01-07 (non-DOM ones; failures were intentional demos: stack overflow, TDZ, uncaught async errors, module `export`s); vitest+jsdom with react 19.3 for React claims; tsc 5.9 for ref typing; curl/python fetches of react.dev, MDN; HTTP check of every "unverified" link. Scratch: tools/sandbox/scratch/p0p1-a/.

01 How the browser works

- Long Tasks API availability: MDN says limited availability, confirmed, hedge replaced with dated statement. Corrected wording.
- `scheduler.yield()`: MDN "limited availability / not Baseline", confirmed; rewrote unverified note.
- 3 unverified links (MDN How browsers work, Chrome inside-browser): HTTP 200, markers removed.
- Process-model teaching model: kept (portable claims only). Confirmed.

02 HTML parsing / DOM / CSSOM

- React 19 head hoisting/stylesheet precedence: consistent with react.dev; left as general statement (not run). Confirmed (docs).
- `<Activity>`: was "volatile"; **corrected**: stable in React 19.2 (blog 2025-10-01); hidden mode unmounts effects (contrast with `hidden` attribute).
- D2 hydration mismatch (`<p><div>`): **executed** (renderToString + jsdom + hydrateRoot): jsdom reparses to `<p></p><div>x</div><p></p>`, React emits nesting warning and recoverable error, ends with client-rendered DOM. Confirmed; detail added.
- Links: 4 unverified MDN/spec links all HTTP 200; markers removed.

03 Rendering pipeline

- `field-sizing`: MDN now Baseline newly available (June 2026); updated.
- Ref callback cleanup (React 19) executed: `attach0,cleanup0,attach1`; confirmed. TS claim that returning a non-function value from a ref callback errors: tsc with @types/react 19.3 gives TS2322; confirmed.
- CSS triggers table, anchor positioning, View Transitions, useEffect-after-paint: hedged heuristics left as is; **unresolved** (not executable; engine-dependent). Anchor positioning support not re-checked.
- web.dev link HTTP 200.

04 DOM structure and mutation

- React 17+ root-container event delegation: confirmed (documented behaviour; not executed).
- Focus/selection on moves, LIS reorder explanation: standard; **unresolved** for browser-specific focus-on-move (not run, hedge retained).
- Chart `useEffect([])` example: type-only; no change.

05 Execution contexts and call stack

- Code samples executed: TDZ/hoisting, recursion RangeError, async ordering (1, A, B), trampoline 1M: all behave as stated. Confirmed.
- Tail-call claim (only JSC implements PTC): consistent with known state; not re-fetched. Left hedged.
- Scheduler/MessageChannel note: matches React scheduler design; "verify file names" marker removed, links 200.

06 Lexical scope and closures

- Closure demos executed: `2 1`, `[3,3,3]` (var) vs `[0,1,2]` (let). Confirmed.
- `useEffectEvent`: was "volatile/verify"; **corrected**: stable in 19.2, not in 19.0/19.1, Effects-only, intentionally non-stable identity, must not be in deps (react.dev reference). All ⚠ on it removed (06 and 07, incl. tables).
- React Compiler: v1.0 stable (blog 2025-10-07); updated.
- Strict Mode effect double-run: documented behaviour; marker removed.

07 Functions and referential identity

- Strict Mode double compute of useMemo: executed (2 calls at mount), confirmed.
- `setState(sameRef)` extra component call before bailout: executed (component function ran again, no commit change), confirmed; marker removed.
- Ref callback identity/cleanup: executed, confirmed.
- React Compiler rows updated to v1.0 stable; `useEffectEvent` rows now say React 19.2+.
- `memo` wrapper-vs-fiber ⚠ note kept (teaching model).

Unresolved

- Browser-specific compositor/layer-promotion claims (03), focus retention on node moves (04), anchor-positioning support (03), PTC in Safari (05). All already hedged in-text.

p0p1 part B: 00-web-and-javascript-foundations lessons 08-14

Environment: Node 24.16, React/ReactDOM 19.3.1, jsdom, Vitest 3.2.7. Scratch: tools/sandbox/scratch/p0p1-b/ (f1.js, b2.js, b4.js, e.js, m/*.mjs, r.test.tsx, fs.test.tsx).

08 Objects, references, immutability

- structuredClone behaviour (Date/Map/Error/cycles cloned; functions and symbol values throw DataCloneError; class prototype dropped; getters flattened): node, CONFIRMED. Added the checked detail.
- Object.freeze shallow / TypeError in strict mode: node, CONFIRMED.
- toSorted / with exist (Node 24): CONFIRMED.
- Immer auto-freeze "in development": immerjs docs say produce output is frozen (not dev-only). CORRECTED (two places).
- React Compiler note: hedged wording only. Unverified anchors in resource table: UNRESOLVED (not fetched).

09 Prototypes and object model

- Class components still supported, error boundaries need class methods: react.dev/reference/react/Component, CONFIRMED; markers removed.
- isReactComponent marker exists in react@19.3.1 source: CONFIRMED.
- Lost `this` on destructured method (undefined in class bodies), #x in o, Object.hasOwn: node, CONFIRMED.
- "No hook equivalent for error boundaries": docs still show only class API; HEDGED.
- useEffectEvent: now stable (19.2), updated.

10 Modules

- Cycle B1 (ReferenceError TDZ), B2 (function hoisted: `b sees a = [Function: a]`, then `a sees b = B`), B3 (`2 2`), assignment to import = TypeError "Assignment to constant variable.": node ESM, CONFIRMED.
- React.lazy caches rejection: React 19.3.1 test, factory called once across two mounts, CONFIRMED; wording made definitive.
- Bundler used by Vite: Vite 8 (npm latest 8.3.5) uses Rolldown; CORRECTED/updated note.
- RSC/Fast Refresh/tool-specific claims: UNRESOLVED (left hedged).

11 Event loop

- 4.1 / B1 `A E G C D F B`, B2 `5 2 3 1 4`: node, CONFIRMED.
- 4.2 `handler end` then `render 2`: Vitest+jsdom, CONFIRMED.
- Flush mechanism: React 19.3.1 source shows sync-lane update scheduled via scheduleMicrotask/processRootScheduleInMicrotask; passive effects of SyncLane flushed at end of commit. CORRECTED (replaced "implementation detail may be end of dispatch or microtask" with the microtask fact).
- flushSync in useEffect: console error text confirmed, and work is deferred (DOM unchanged on next line). CORRECTED lesson to quote the real message and behaviour.

- useEffectEvent stable in 19.2 (react.dev blog 2025-10-01): CONFIRMED.
- scheduler.yield / rIC / MessageChannel / B3 real-click vs script click semantics: not run (needs browser). UNRESOLVED (kept hedges).

12 Promises

- 3.2 `a1 b1 a2 b2 a3`, 4.1 chain output (1,2,x,fin,recovered), 4.2 `a1 b1 sync end a2 b2 a3`, B2, B3, B4 (a rejects, b caught, c "no error" + unhandled), F1 (`a1 b1 p1 p2 a2 p3`; plain-value variant `a1 b1 a2 p1 p2 p3`): node, all CONFIRMED.
- async transition post-await caveat and `use` promise-stability pitfall: react.dev useTransition/use pages, CONFIRMED. Markers removed.
- AbortSignal.any/timeout, Promise.withResolvers exist in Node 24: CONFIRMED (browser support left hedged).

13 Browser events

- 4.2 order `native document capture, a capture, c bubble c, a bubble, native root bubble, native document bubble` and stopPropagation variant: Vitest+jsdom, CONFIRMED.
- React registers wheel/touchstart/touchmove as passive: spied addEventListener, CONFIRMED.
- onFocus bubbles (focusin): CONFIRMED. onChange per input event: not run.
- 5.4 "menu flashes open and closes": NOT reproducible with scripted click (no microtask checkpoint). CORRECTED: explained that it needs a real user click; mechanism text fixed (microtask flush).
- Capture-before-bubble at target (DOM spec change): UNRESOLVED, not tested (jsdom would not be evidence); hedge kept.
- Disabled-button click behaviour: jsdom `.click()` does not dispatch; browser differences UNRESOLVED.

14 Web APIs

- Strict Mode effect: start, abort, start: CONFIRMED (B5 answer holds).
- Node: no default localStorage in v24, ReadableStream async iterator present, AbortSignal.any present, aborted fetch rejects AbortError, custom reason propagates: node, CONFIRMED/CORRECTED text.
- Network-panel statuses, priority option, bfcache timer note: UNRESOLVED (browser-only).

p0p1 part C: Phase 0 lessons 15-21 (reviewer changelog, 2026-10-22)

Scope was narrowed to risky, claim-bearing items. Most prose and browser/network claims in 15-17 were reviewed for plausibility but not machine-checked.

Executed / sources

- tsc 5.9.3 + @types/react 19.3.0: scratch at tools/sandbox/scratch/p0p1-c/ (a.tsx; b/ holds every ts/tsx block from lessons 18-19 extracted and compiled).
- npm view vite (8.3.5), Vite 8 announcement and guides, react.dev CRA sunset post, TS erasableSyntaxOnly (tsc flag run), node 24 (AbortSignal.any/timeout exist).

Lesson 15 (HTTP/caching)

- vite:preloadError event name: confirmed (Vite build guide). Marker removed.
- Cache-Control/bfcache/103/Navigation API/SW claims: plausible, not verified. **Unresolved**; markers kept.

Lesson 16 (memory/GC): no changes; Strict Mode effect claim consistent with React docs. Engine-detail hedges left. Confirmed-by-inspection only.

Lesson 17 (perf)

- INP good <= 200 ms at p75: consistent with web.dev (page states p75 methodology). Left hedged. V8 tier names, Long Tasks support, scheduler.yield: **unresolved**, markers kept.

Lesson 18 (TS)

- filter(Boolean) not narrowing; filter(x => x !== undefined) narrows (5.5): confirmed by tsc. Marker removed.
- TS 5.4 closure narrowing after last assignment: confirmed (later reassignment re-widens). Markers removed.
- Remaining code blocks compile except intentional-error exercise snippets and fragments with missing declarations (Shape exhaustiveness fragment, stubs).

Lesson 19 (TS generics/React types)

- useRef() requires argument, global JSX namespace removed, ReactElement props unknown, `<Ctx value>` and `<Ctx.Provider>` both valid: confirmed. Markers removed/updated.
- forwardRef/ref-as-prop, Strict Mode request count: consistent with React 19 docs; not run. Left.

Lesson 20 (tooling) -- CORRECTED

- Vite described as esbuild-dev + Rollup-prod with Rolldown "in development": outdated. Vite 8 (current 8.3.x) ships Rolldown as the single bundler, Node ^20.19||>=22.12. Added version note, updated pipeline steps and comparison table.
- CRA deprecation: confirmed (react.dev, 2025-02-14). erasableSyntaxOnly in TS 5.8+: confirmed (flag works in 5.9.3).
- Dev proxy `ws: true`, manualChunks Rolldown compat: **unresolved**.

Lesson 21 (assessment): removed marker on TS 5.5 inferred predicates (confirmed). Other answer keys not re-derived.

Totals: confirmed 10, corrected 1 (plus 3 hedge/version updates), unresolved ~8 (browser/HTTP-level claims).

p0p1 part d: 01-react-fundamentals lessons 01-10 (reviewed 2026-10-22)

Scripts: /Users/todorbra/Projects/ReactLearning/tools/sandbox/scratch/p0p1-d/ (a,b,c,d .test.tsx via vitest+jsdom, jsx.mjs via TS 5.9). React 19.3.1.

Scope note: I targeted the risky/version-sensitive claims in each lesson and did not read every line of all 10 lessons.

- 01: StrictMode double render, no double in plain (b.test): confirmed. Version claims on render/hydrate removal and forwardRef: confirmed via upgrade guide/sandbox. Unresolved: exact release dates, react-reconciler API stability (left hedged).
- 02: StrictMode effect mount/cleanup/mount: confirmed (`A eff, A cleanup, A eff`).
- 03: batching in setTimeout = 1 render; same-value setState bail-out "may render once more": confirmed (one extra render then none). Unresolved: APG tabs URL (unfetched).
- 04: `$$typeof` = Symbol(react.transitional.element), element keys, frozen element/props, assignment throws TypeError: confirmed. CORRECTED: `element.ref` is removed (dev warning, still returns value), not "deprecated". `ref` in props for function components: confirmed. Unresolved: @types/react return-type notes, Dan Abramov URL.
- 05: new JSX transform required in 19: confirmed. key after spread falls back to createElement: confirmed with TS 5.9 (jsx and jsxDEV output). Key-spread dev warning: confirmed. CORRECTED: javascript: URLs are replaced with a throwing stub, not just warned. bigint child renders: confirmed. Promise child: unresolved on client (sandbox SSR suspends; hedge kept).
- 06: defaultProps removed for function components (renders "undefined"): confirmed. props.key read warns: confirmed. frozen props: confirmed.
- 07: children-prop bail-out (Wrapper only re-renders): confirmed. CORRECTED/verified Children.map empty nodes (callback called for false/null, index counts them; toArray drops). Context as provider `<C value>` : confirmed.
- 08: undefined return renders nothing: confirmed. Activity exists in 19.3.1, hidden cleans effects, keeps state: confirmed (marker removed).
- 09: key not a prop, spread-key warning: confirmed. toSorted/groupBy runtime support: unresolved (left as hedged).
- 10: SyntheticBaseEvent, nativeEvent, currentTarget null after dispatch, persist exists, capture/bubble ordering: confirmed. Passive touchstart/touchmove/wheel: confirmed in source (marker removed). useEffectEvent exists in 19.3.1 react (exported); status left hedged. Form actions: not executed (unresolved).

p0p1 part E: 01-react-fundamentals lessons 11-20 (reviewed 2026-10-22)

Scratch: tools/sandbox/scratch/p0p1-e/ (a-e.test.tsx, ts/, cv/ = real create-vite scaffold). React 19.3.1, jsdom, vitest 3.

Executed checks (React 19.3.1, createRoot + act)

- L12 5.1 impure counter: StrictMode shows 2 / 4; without StrictMode 1 / 2. CONFIRMED (added "verified").
- L12 3.1 / L17: Strict Mode order on mount `init,init,memo,memo,render,render,ref attach,setup,ref detach,cleanup,ref attach,setup`. Updater runs twice (`upd,upd`). useMemo factory runs twice total per render (reuse across the double render consistent with the React 19 upgrade guide). CONFIRMED; wording corrected (L12 3.1, L17 2.x: ref-callback re-run is documented on react.dev StrictMode).
- L13: `<Profiler onRender>` under StrictMode: mount + 1 click = 2 calls vs 4 component calls. CONFIRMED. react.dev/Profiler: disabled in production by default. CONFIRMED.
- L13 console dimming: React does not dim; DevTools does ("Hide logs during additional invocations in Strict Mode" setting exists). CORRECTED wording.
- L11 3.6 / L11 D4: after `<form action={fn}>` completes, uncontrolled defaultValue input reverts, controlled input keeps state value. CONFIRMED (marker removed, evidence added).
- L11 "Stuck" A/B warn, C/D editable: 2 console.error warnings for A, B. CONFIRMED.
- L10 3.3 ordering `React outer capture > native root > native inner > React inner bubble > React outer bubble > native document`. CONFIRMED. React stopPropagation does not stop native inner listener. CONFIRMED. onFocus bubbles; onScroll does not. CONFIRMED. touchstart/touchmove/wheel registered passive at root. CONFIRMED.
- L15: 84 result, n = 1 then 4 (updater queue), automatic batching in setTimeout (1 render), updater double-invocation in Strict Mode. CONFIRMED.
- L16: same-type/same-position preservation, different parent type resets state, nested component definition remounts input (new DOM node each keystroke). CONFIRMED. `<Activity>`: hidden children get `display: none !important`, state kept, effects cleaned up on hide and re-setup on show. CONFIRMED (stable since 19.2; markers removed).
- L17: effect order A,B,C mount / A,C on click; child effect before parent effect; Strict Mode effect cycle ABAB; leaky interval ticks 2/s in dev and keeps ticking after unmount. CONFIRMED.
- L14: `useEffectEvent` exported as function in react 19.3.1; react.dev reference page exists. CORRECTED (was "experimental/verify status"; now stable since 19.2).
- L19: snippet typechecked with tsc 5.9 strict; `step="2"` gives TS2322. CONFIRMED.
- Links: all 66 URLs in lessons 11-20 fetched with curl: HTTP 200 (except api.example.com placeholder and npmjs.com 403 bot block). "(unverified)" tags removed.

L19 Vite setup: CORRECTED (largest drift)

Scaffolded `npm create vite@latest hello -- --template react-ts` (create-vite 9.3.1, Node ^20.19 || >=22.12): Vite ^8 (Rolldown), @vitejs/plugin-react ^6, TypeScript ~6.0, oxlint (`.oxlintrc.json`, `"lint":

`"oxlint"`, no eslint.config.js, no vite-env.d.ts, public/favicon.svg+icons.svg), tsconfig target es2023, erasableSyntaxOnly. Lesson updated: file tree, lint script, plugin-react description, bundler statement, ESLint section now framed as the earlier-template/optional route; react-hooks preset changed to `configs.flat.recommended` (v7 README). Note: TS 6.0 in template, TS 7.0 exists (not covered here).

L18 DevTools

Verified from DevTools source/CHANGELOG (8.0.0, Sep 2026): label "Highlight updates when components render", "Record why each component rendered while profiling" (Settings > Profiler), "Always parse hook names from source". Added: Suspense tab on by default, Timeline tab removed in 8.0. Bug-icon label left hedged (could not locate). Not run in a real browser (no extension in sandbox): UI walkthrough steps UNRESOLVED as to exact current layout.

Marker cleanup

Removed all remaining warning markers (⚠) in lessons 10-20 (10 included, assessed in same batch scope for L10). Where claims were not individually executed, existing prose hedges ("verify", "version-sensitive") remain.

Unresolved / not executed

- Hydration-mismatch behaviour in React 19 (L12 D2/error reporting): not executed (no SSR hydration run); prose kept, left hedged.
- IME composition behaviour with controlled inputs, masked-input caret (L11): jsdom cannot reproduce; unresolved.
- Fast Refresh state-preservation (L19): not run in a browser.
- React Compiler claims (L12/L13): only checked that Compiler 1.0 is stable (react.dev blog) and babel-plugin-react-compiler 1.0.0 exists; not run.
- Newer-version note: React 19.3 (View Transitions, Fragment Refs) exists; not relevant to these lessons, no note added.